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Climate resiliency assessment of drinking water technologies in southwest coastal Bangladesh using the 4R framework

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Abstract

Climate change is intensifying hydro-climatic extremes and exacerbating drinking water insecurity in coastal regions. In southwestern coastal Bangladesh, progressive salinisation combined with recurrent disasters threatens safe water access. This study assessed the climate resiliency of six existing drinking water technologies (DWTs), namely community-based rainwater harvesting (CRWH), deep tube wells (DTWs), pond sand filters (PSFs), solar-powered PSFs (SPSFs), desalination plants (DSPs), and ultrafiltration plants (UFPs), in the Bagali Union of Koyra Upazila under Khulna District of Bangladesh. The 4R (robustness, redundancy, resourcefulness, and rapidity) framework was conceptualised through a multicriteria analysis (MCA) to assess the climate resiliency of the DWTs. Data were collected using a mixed-method approach integrating water quality analysis and household surveys with focus group discussions and key informant interviews. Water quality results indicated that DSP delivered the highest quality water, followed by CRWH and UFP, whereas DTW showed excessive salinity ($EC = 2601 \mu S/cm$), far beyond the permissible limit. The resiliency ranking differed, where CRWH was the most resilient (0.246), followed by SPSF (0.180), DSP (0.159), PSF (0.149), DTW (0.139), and UFP (0.127). The CRWH benefitted from community participation, active maintenance, and long operational life, whereas DSP and UFP were constrained by energy dependency, high operational cost, and limited local repair capacity. Expanding CRWH within a complementary technology portfolio and strengthening governance mechanisms can significantly increase the climate-resiliency of drinking water supplies in similar hydrogeological and institutional settings across coastal Bangladesh.

Keywords Climate resiliency, Drinking water technologies, 4R framework, Multicriteria analysis, Gender and water access, Coastal Bangladesh

1 Introduction

Safe drinking water is recognised as a fundamental human right, yet its availability is increasingly threatened worldwide as climate change alters hydrological processes and water resource systems [1, 2]. Climate-driven water insecurity manifests through altered precipitation regimes, intensified evapotranspiration, and increased frequency



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of extreme events, including floods, droughts, and storm surges, that simultaneously reduce freshwater availability and degrade existing water quality [3]. Approximately four billion people face severe water scarcity for at least one month per year, with this figure projected to rise substantially under warming scenarios [4]. Deltaic and low-lying coastal regions are disproportionately affected, where the confluence of sea-level rise, salinity intrusion, land subsidence, and high population density creates compounding and persistent threats to drinking water security [5, 6].

Bangladesh exemplifies these intersecting vulnerabilities at a critical scale. The southwestern coastal zone of the country, identified as a hard-to-reach area encompassing the Khulna, Satkhira, and Bagerhat districts, stretches across approximately 12,206 km² and hosts more than 6.42 million people [7, 8]. This region is experiencing progressive salinisation of both surface water and groundwater, driven by reduced upstream freshwater flows, embankment failures, commercial shrimp aquaculture, and cyclonic storm surges that have penetrated more than 100 km inland [9]. During the dry season (November–May), the average surface water salinity in affected areas exceeds the permissible limit, indicating that the majority of water sources are unsafe for consumption. Groundwater-based solutions are equally constrained, as coastal aquifers are dominated by highly saline water, often exceeding 2000 µS/cm [10]. The hydrogeological context thereby fundamentally limits the suite of feasible drinking water technologies (DWTs) and makes DWT selection in coastal Bangladesh one of the most consequential water sector decisions in South Asia.

The public health consequences of deteriorating drinking water quality in this region are well documented. The consumption of high-salinity water has been strongly associated with elevated blood pressure among adults, as well as with hypertensive disorders of pregnancy, including gestational hypertension, and pre-eclampsia, with hospital admissions for maternal hypertensive crises rising during the dry season [11–14]. A recent meta-analysis pooling 27 observational studies across seven countries and 74,063 participants reported a mean systolic blood pressure elevation of +3.22 mmHg and a pooled odds ratio of 1.26 for hypertension among populations exposed to elevated drinking water salinity. These findings establish drinking water salinity as a robust and quantifiable cardiometabolic risk factor [15]. In addition to maternal health, salinity inflicts substantial agricultural and economic losses, estimated in billions of USD annually, reinforcing cycles of poverty that further limit household adaptive capacity [16, 17]. In this context, the reliability and resiliency of DWTs transcend technical considerations to become matters of survival and development equity.

In response to these escalating challenges, a diverse range of DWTs have been introduced in the coastal areas of Bangladesh through government programmes, non-government organization (NGO) initiatives, and international development assistance [18]. Technologies range from low-technology, community-managed systems such as rain-water harvesting and pond sand filtration to capital-intensive desalination and ultra-filtration plants [19]. However, water sector investments have historically prioritised installation coverage over long-term functionality, and a consistent pattern across South Asia and sub-Saharan Africa shows that 30–50% of installed community water points become non-functional within a few years of installation [20, 21]. In coastal Bangladesh specifically, cyclonic events such as *Aila* (2009), *Amphan* (2020), and *Remal* (2024) have

repeatedly incapacitated many of the installed DWTs, demonstrating that even newly installed infrastructure may fail precisely when it is most critically needed [22–24].

This widespread infrastructural failure points to a fundamental conceptual limitation in how DWT performance is conventionally assessed. Physical installation and even momentary water quality compliance are insufficient proxies for what communities truly need, which is a resilient system that continues to deliver safe water before, during, and after climate shocks. Resiliency, defined as the capacity of a system to absorb disturbances, adapt, and continue delivering essential services [25], provides a more comprehensive evaluative lens. The 4R (robustness, redundancy, resourcefulness, and rapidity) framework of Bruneau et al. [26] decomposes resiliency into robustness (the ability to withstand stress without loss of function), redundancy (the substitutability of system elements), resourcefulness (the capacity to identify problems and mobilise resources), and rapidity (the speed of recovery to a desired level of functionality). It builds on the earlier reliability–resiliency–vulnerability metrics of Hashimoto et al. [27], which remain foundational for water-system performance evaluation. Although the framework was developed for the seismic resiliency of urban infrastructure, it has since been extended to water systems [28], water-supply disaster response [29], disaster risk management [30], and climate adaptation planning [31]. Its systematic application to rural DWTs in climate-vulnerable developing country settings nevertheless remains limited.

Research on drinking water in coastal Bangladesh has largely focused on single dimensions of system performance, including water quality compliance [32, 33], hydrogeological feasibility [10, 34], or the social dimensions of technology adoption [35, 36]. Additionally, some studies simultaneously consider the technical suitability, economic viability, and social acceptability of DWTs [37, 38]. However, evaluations of technologies that consider technical performance, management, maintenance capacity, and disaster recovery within a unified resiliency framework are scarce. Furthermore, comparative assessments spanning the full range of co-existing DWT types, from traditional rain-water harvesting to advanced membrane filtration, are largely absent from the peer-reviewed literature. Butler et al. [28] demonstrated that reliable, resilient, and sustainable water management requires integrating all three dimensions simultaneously, whereas Whittington et al. [40] found that rural water system failures are more often rooted in weak governance and inadequate maintenance than in purely technological limitations. This leaves water sector planners and policymakers without the evidence base needed to make informed investment decisions that account for climate trajectories and community-level resiliency requirements [42].

This study addresses these gaps by assessing the climate resiliency of six co-existing DWTs in the Bagali Union of Koyra Upazila under Khulna District, a community in the heart of the hard-to-reach coastal zone of Bangladesh. The study integrates hydro-lithological characterisation, water quality testing, household surveys, and qualitative insights within a multicriteria analysis (MCA) framework grounded in the 4R resiliency dimensions. The specific objectives of the study are (i) to map existing DWTs and dependent households; (ii) to examine the water quality of the DWTs during the peak dry season; and (iii) to assess the climate resiliency of the DWTs. The findings aim to redirect investment logic from coverage-based infrastructure expansion toward resiliency-grounded service delivery, a transition essential for achieving Sustainable Development Goal 6.1 in the world's most climate-threatened deltaic communities [41].

2 Study area

The study was conducted in the Bagali Union of Koyra Upazila under Khulna District, in the southwestern coastal region of Bangladesh (Fig. 1). The Union encompasses 29 villages across approximately 49 km², with low-lying topography and acute exposure to hydroclimatic hazards. The region experiences a tropical monsoon climate with a pronounced wet season (June–October) and a subsequent dry season (November–May), during which salinity levels peak and water scarcity intensifies [42, 43].

The Bagali Union was purposively selected because of its severe and multidimensional water insecurity and the coexistence of diverse DWTs under the same hydrogeological and institutional setting. The Union has experienced repeated catastrophic impacts from tropical cyclones, most notably Cyclone *Aila* (2009), *Amphan* (2020), and *Remal* (2024), which caused prolonged salinisation of surface and groundwater sources,

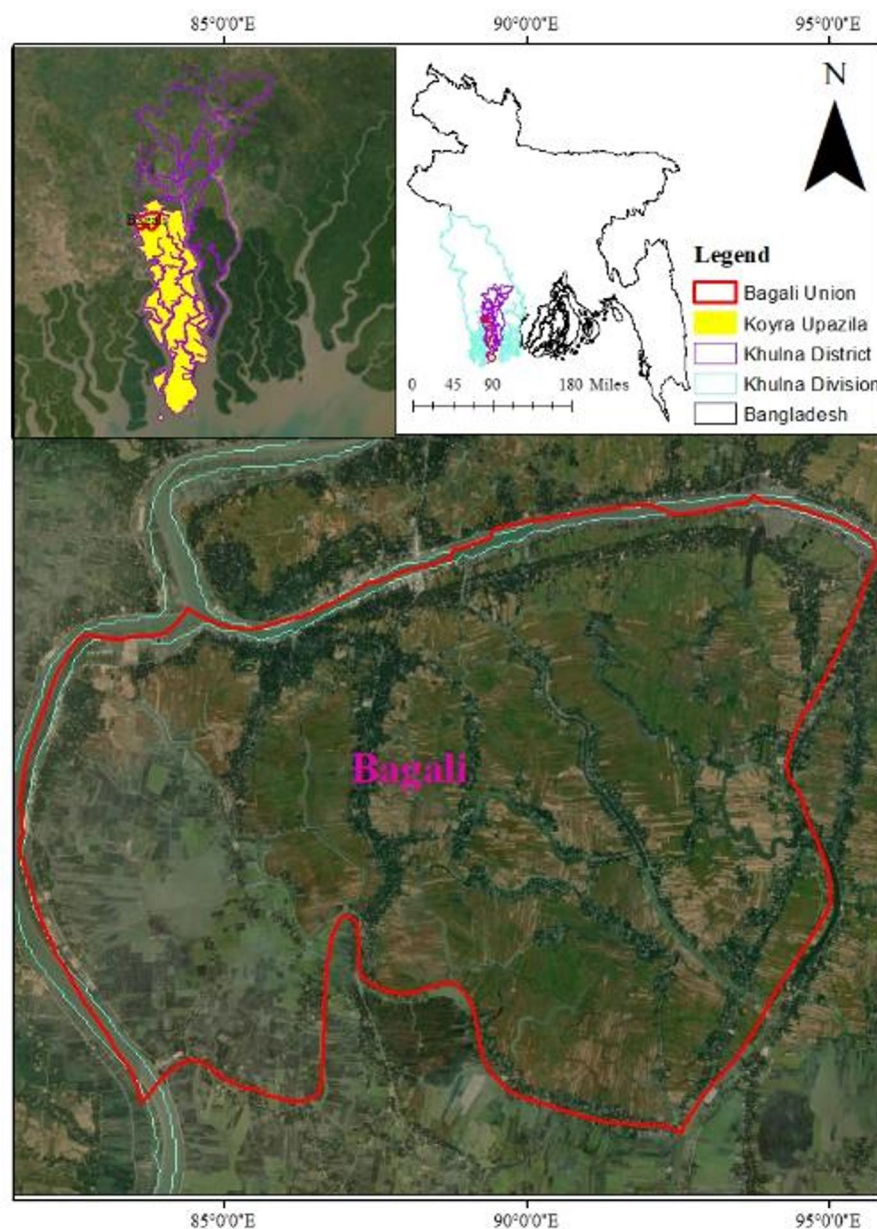


Fig. 1 Map of the study area (the Bagali Union of Koyra Upazila under Khulna District)

damaged embankments, and incapacitated the majority of installed water infrastructure for extended periods [24]. This legacy has shaped the current DWT landscape, triggering successive rounds of rehabilitation and technology introduction. The Department of Public Health Engineering (DPHE), the Local Government Engineering Department (LGED), and NGOs have consequently introduced six improved DWTs, namely community-based rainwater harvesting (CRWH), deep tube wells (DTWs), pond sand filters (PSFs), solar-powered pond sand filters (SPSFs), desalination plants (DSPs), and ultrafiltration plants (UFPs). These were intended to reduce dependence on traditional sources such as open pond water and household rainwater storage [23]. These technologies differ in water source, energy requirement, ownership model, and maintenance requirements, yet now operate side by side within the Union.

3 Materials and methods

3.1 Conceptual framework

The climate resiliency of a DWT is defined as its capacity to continue delivering safe drinking water before, during, and after climate shocks. Resiliency was assessed via the 4R framework [26], which was modified for the rural coastal water sector context, drawing on recent water-system-specific 4R operationalisations [29] and the broader review of drinking water resiliency frameworks by Charles et al. [31]. The 4R framework covers twelve criteria informed by the technical, organisational, social, and economic (TOSE) dimensions of Butler et al. [28] and adapted from similar existing studies in south-west coastal Bangladesh [37, 38] and iterative stakeholder consultation. Robustness criteria included water quality, operating life, and structural resistance to disasters. Redundancy criteria included source substitutability, availability of spare parts, and support systems (active committee). Resourcefulness criteria included gender roles in management, operation, and maintenance (O&M) capacity and ease of repair (availability of skilled technicians). Rapidity criteria included accessibility during and after disasters, emergency response time, and social support for restoration. All these 4R criteria were identified from the initial field survey and the literature [9, 37]. The framework, which maps TOSE dimensions onto the 4R criteria for each DWT type, is presented in the Supplementary Material (Fig. S1).

3.2 Study design and sampling

A mixed-method research design was employed, coupling quantitative water quality data and structured household surveys with qualitative evidence. This design was selected to capture the multilayered determinants of DWT resiliency, which cannot be adequately assessed through any single method [44]. Fieldwork was conducted during the dry season (March–May 2025), when salinity stress and water scarcity are most acute and differences in DWT performance are most pronounced. A reconnaissance survey (October 2024) preceded the main fieldwork to enumerate DWT sites, assess functional status, geo-reference units, and establish stakeholder contacts.

Household surveys ($N=180$) were conducted via a semi-structured questionnaire. The respondents were selected through a stratified random sampling design, in which the six DWT types formed the strata and 30 user households were selected from each, including both men and women so that both perspectives were captured. Households using more than one technology seasonally were assigned to the stratum of their primary dry-season

source, ensuring that no household appeared in more than one user group. Because allocation was equal rather than proportional to the number of dependent households. The survey captured DWT use patterns and seasonal switching, perceived water quality, household expenditures on water, access distance and travel time, experiences of and recovery from climate-related disruptions, and participation in DWT management. The questionnaire was pretested with 15 respondents in the Union and refined accordingly; these pretest responses were excluded from the final results.

Six focus group discussions (FGDs) were conducted, one per technology, with 8–10 participants each. Participants were selected to ensure gender balance and representation across villages. The FGD guides explored community governance structures, maintenance challenges, gender roles, disaster coping strategies, and willingness to pay for improved service. Twelve key informant interviews (KIIs) covered DWT owners and managers ($n=6$) on operational and financial aspects and elected representatives, government officials, and NGO representatives ($n=6$) on governance and policy. Findings were triangulated across the three evidence streams. Converging themes were treated as core findings, whereas divergent accounts were interpreted with reference to the informant's position in the system rather than resolved by majority. Where accounts diverged, the more conservative reading was adopted when assigning criterion scores. Requiring agreement across independent sources before a finding informs the analysis is the principal safeguard against response and interpretation bias.

3.3 Water quality analysis

Water samples were collected from all six DWT types during the peak dry season (March–May 2025), when salinity stress and performance differences are greatest. Seventeen samples were collected, three from each of five technologies and two from UFP. Sampling units were functional installations identified during the reconnaissance survey. They were selected to span the geographic extent of each technology's catchment and were drawn from different villages wherever more than three units were available, so that the samples reflect between-unit variation. One of the three UFP units was non-functional during the final survey, which accounts for the shortfall. Each sample is a single grab sample from a distinct installation, so the values per technology are field replicates across units rather than analytical replicates of one unit.

On-site water testing of pH, electrical conductivity (EC), and total dissolved solids (TDS) were performed using a HACH multimeter calibrated against standard solutions at the start of each sampling day. Samples for laboratory analysis were collected in pre-sterilised bottles and transported to the Soil, Water, and Environment Discipline Laboratory, Khulna University, within 6 h. All analyses were completed within 24 h of collection, within the recommended holding times [45].

Laboratory analyses followed standard methods [45]: faecal coliform (FC) isolation by membrane filtration with mFC incubated at 44.5 °C for 24 h [46]; total nitrogen (TN) by Kjeldahl digestion and distillation [47]; and total phosphorus (TP) by the vanadomolybdophosphoric acid yellow colour method with spectrophotometric measurement at 470 nm [48]. These three parameters were measured only for the surface-water-based technologies (CRWH, PSE, SPSE, and UFP) and not for the groundwater-based DTW or DSP, because deep and confined aquifers are hydraulically separated from surface faecal and nutrient pathways in this setting [10]. Results were compared against the Bangladesh

national standards in the Environmental Conservation Rules (ECR) [49] and the World Health Organization (WHO) guidelines [50], noting that the ECR prescribes no limit for EC in drinking water.

3.4 Spatial and lithological analysis

The spatial distributions of functional and non-functional DWTs and household catchments were mapped using ArcMap 10.8. Lithological and stratigraphic profiling was performed via RockWorks 17 software to characterise subsurface aquifer conditions, integrating borehole log data from local DPHE records. Three borehole logs were available within the Bagali Union, logged to a maximum depth of 260 m below ground level in March 2022.

3.5 Multicriteria Analysis

The six DWTs were ranked by MCA using the weighted sum method (WSM). This method is widely used in environmental decision-making for its transparency and suitability for ordinal scoring [51] and has been applied to coastal water supply in Bangladesh [38, 39] and to resiliency-based utility ranking and resilient–sustainable water management elsewhere [52–55]. Each criterion was scored 3 (high), 2 (moderate), or 1 (low) for each technology by the author team, with every score required to rest on triangulated field evidence. Two decision rules were applied uniformly. For criteria with a measured basis, the six technologies were ordered on the relevant field quantity and the distribution was divided into three bands. Water quality was banded on the aggregate compliance score in Table S4, and the remaining measured criteria on the household survey findings reported in Sect. 4.4. For qualitatively grounded criteria, principally gender roles in management and social support, a category was assigned only where FGD and KII accounts converged, with the more conservative reading adopted where they did not. The complete score matrix is given in Table S5.

Equal weights were applied, with each 4R dimension assigned 25% and each constituent criterion assigned $0.25/3 = 0.083$. Equal weighting was chosen on substantive grounds. In this setting, the four dimensions are jointly binding rather than substitutable. Failure in any dimension interrupts water supply regardless of performance in the others. Therefore, there was no basis for assigning greater weight to one dimension over another. Raw scores were normalised within each criterion before weighting. The normalised performance of technology i on criterion j is

$$r_{ij} = \frac{S_{ij}}{\sum_{k=1}^6 S_{kj}} \quad (1)$$

where S_{ij} is the raw score of technology i on criterion j , and the denominator sums the raw scores of all six technologies on that criterion. The composite resiliency score is then

$$R_i = \sum_{j=1}^{12} w_j \cdot r_{ij} \quad (2)$$

with $w_j = 0.083$. On the water quality criterion, for example, raw scores are CRWH 3, DTW 2, PSF 1, SPSF 1, DSP 3, and UFP 2, summing to 12, so the weighted contribution of CRWH is $0.083 \times (3/12) = 0.021$, the value shown in Table 3.

Because $\sum j r_{ij} = 1$ for every criterion, the six composite scores sum to approximately 1.00. They are therefore relative shares of the total assessed 4R performance across the six alternatives, not independent absolute indices, and are meaningful only in comparison with one another against an equal-share benchmark of $1/6 = 0.167$.

3.6 Limitations

The study covers a single, purposively selected Union, so the findings characterise the Bagali case and are not statistically generalisable, although the area shares the defining hydro-climatic and institutional conditions of much of southwest coastal Bangladesh. Data were collected during the peak dry season, so the water quality results describe conditions of maximum salinity stress and cannot establish year-round quality. Scores were agreed by discussion among the authors based on triangulated field evidence; because a single consensus set was produced rather than independent parallel ratings, no inter-rater agreement statistic could be computed.

4 Results

4.1 Socio-demographic characteristics of the study area

The socio-demographic profile of the surveyed households ($N = 180$) across the six DWT user groups provides an essential context for interpreting drinking water use patterns and resiliency outcomes in the study area. The gender distribution was nearly balanced, with 52% male and 48% female respondents, which ensured that both perspectives on water management were captured. The age profile indicates a predominantly young-to-middle-aged population, with 71% of respondents aged between 28 and 48 years. These groups were specifically targeted to capture individuals with direct and sustained experience in managing household water needs under climate variability. Educational attainment is relatively low, with 85% of respondents holding primary-to-secondary level education and only 15% having completed secondary school or above. This limited educational profile is relevant to understanding the awareness and adoption of complex DWT management practices.

Occupationally, housewives constitute the largest group (45%), reflecting the predominantly female responsibility for household water collection: 55% of respondents identified women as the primary water collectors, 42% men, and 3% paid collectors. This gendered burden is exacerbated during climate-related disruptions, when collection distances increase, and water quality deteriorates. Other prominent livelihoods include farming (17%), business (16%), and day labouring (15%), indicating diverse but generally modest economic stability. The FGDs reinforced that water collection consumes substantial time for women and girls, often limiting their participation in productive activities. A further notable pattern is the gendered disparity in collection method. Women typically fetch water on foot, whereas men predominantly use vehicles, including vans, bicycles, motorbikes, or other available modes of transport (see Supplementary Material, Table S1).

Technology preference was largely independent of respondent gender. Overall, 65% of respondents identified CRWH as the best DWT, with almost similar proportions among women (69%) and men (62%); no other technology exceeded 14% in either group. Household preferences were shaped less by individual choice than by access. While 97% of CRWH users identified CRWH as the best option, 63% of DSP users and 47% of UFP

users also preferred CRWH despite relying on paid alternatives. Preference in this setting is therefore near-universal, and the binding constraint is proximity to a functioning unit rather than individual choice (see Supplementary Material, Tables S2 and S3).

Household economic vulnerability is shaped less by technology preference than by the accessibility and cost of available water sources. Physical access is itself a major constraint, with 45% of households travelling more than 1 km to collect drinking water, creating a substantial time and physical burden, particularly for women (see Supplementary Material, Fig. S2). Overall, 55% of households paid for drinking water, but payment varied markedly by technology. All UFP and DSP users purchased water. Among paying households, the average monthly expenditure was BDT 607 for DSP users and BDT 353 for UFP users. The financial burden translated directly into household welfare. In addition, 22% of respondents reported reducing other household expenditure to afford drinking water, whereas no non-paying household reported making such trade-offs. Poorer households residing in paid-service catchments therefore incur a tangible welfare cost that is not borne by households in community-service catchments (see Supplementary Material, Fig. S3).

Survey data revealed that 41% of households have experienced drinking water shortages in recent years, typically lasting one to two months of the year. Cyclones, deteriorating water quality, and prolonged dry periods were identified as the main causes (see Supplementary Material, Fig. S4). Among the 36% of households reporting water quality problems, 89% experienced these problems exclusively during the dry season, when salinity peaks and contaminant concentrations rise (Fig. 2). The most frequently reported problem was a salty taste, followed by discolouration and odour.

4.2 Hydro-lithological context and DWT landscape

The subsurface profile of the Bagali Union presents a challenging hydrogeological constraint on drinking water provision. Stratigraphic analysis revealed a subsurface dominated by multiple thick aquitard sequences, with alternating silt-clay layers averaging 164 m in cumulative thickness. The sandy aquifer units are thin, discontinuous, and of limited hydraulic capacity (average thickness 55 m). Critically, groundwater in all productive aquifer units tested in the Union exhibits extreme salinity. The average EC value of DTW is 2,601 $\mu\text{S}/\text{cm}$ is well above the 1,500 $\mu\text{S}/\text{cm}$ limit adopted for drinking water in comparable hydrogeochemical assessments [57, 58], and the corresponding TDS exceeds the national and WHO palatability threshold of 1,000 mg/L [50]. The subsurface cross-section is presented in Fig. 3 (a and b).

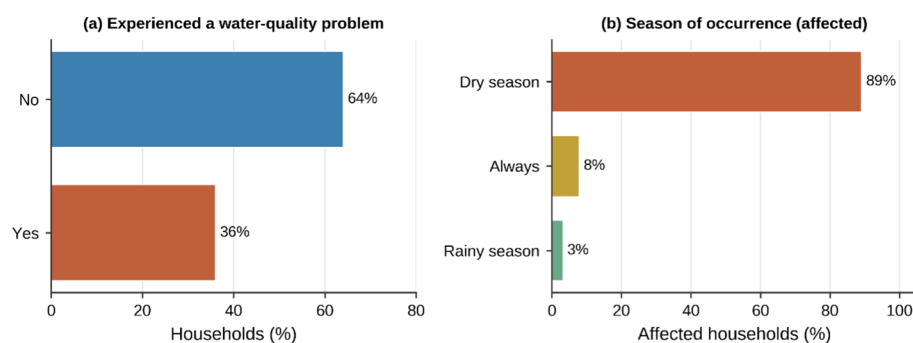


Fig. 2 Experience with water quality problems **a** and the season of occurrence **b** ($N = 180$)

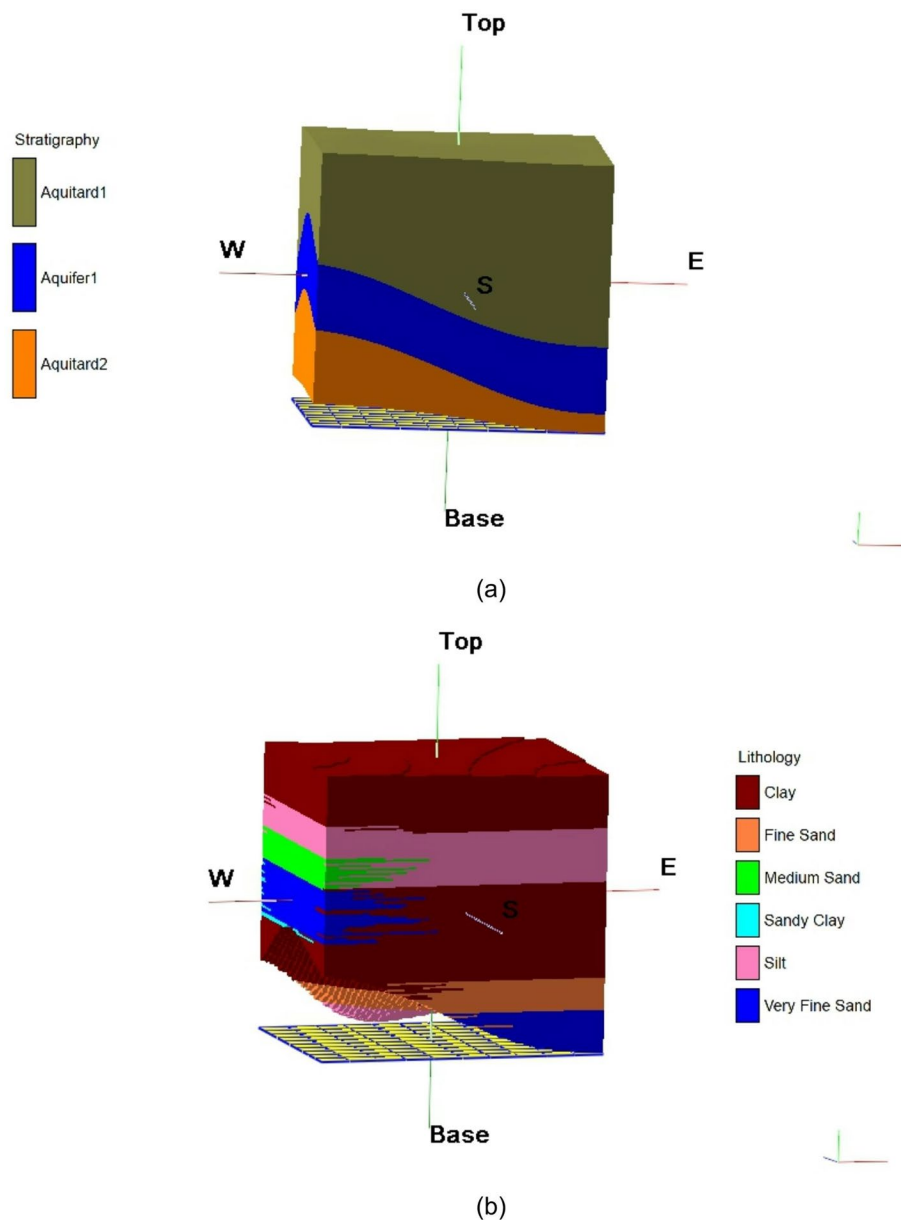


Fig. 3 **a** Stratigraphic profile and **b** lithological cross-section of the study area

This hydrogeological context renders groundwater-based solutions fundamentally unsuitable for potable use across most of the villages of the study area, explaining the coexistence of surface-water-based alternatives. Field enumeration during the reconnaissance survey identified 40 DWT units in the Bagali Union across all six technology types. Of these, 20 units (50%) were functional at the time of the study and 20 (50%) were non-functional. The functional units, together with their catchments and ownership arrangements, are summarised in Table 1. The spatial distributions of functional and non-functional DWTs are presented in the Supplementary Material (Fig. S5). The qualitative findings attributed non-functionality to inadequate post-installation monitoring and maintenance, lack of clear ownership and accountability arrangements, repeated damage from cyclonic flooding and storm surges, and the absence of pre-established maintenance funds.

Table 1 Summary of DWT types, functional status, spatial coverage, and ownership patterns in the study area

DWT type	No. of functional DWTs	No. of villages served	Average No. of dependent households	Maximum access distance (km)	Primary ownership
CRWH	3	7	180	2	Community/Mosque
DTW	3	10	200	>4	Community
PSF	3	8	700	4	Community
SPSF	3	9	900	4	Community
DSP	6	12	2,400	4	Private owner
UFP	2	5	250	>4	Private owner

Table 2 Average measured values of water quality parameters of DWTs compared against the Bangladesh standard. Values in bold exceed the limit

Parameter	Unit	CRWH	DTW	PSF	SPSF	DSP	UFP	ECR standard
pH	–	8.56	7.24	7.38	7.68	6.55	7.66	6.5–8.5
EC	μS/cm	121	2,601	1,202	1,746	378	748	1,500*
TDS	mg/L	60	1,302	593	872	189	358	≤ 1,000
TN	mg/L	13.0	N/A	11.0	25.7	N/A	12.0	≤ 10
TP	mg/L	0.050	N/A	0.090	0.050	N/A	0.110	0
FC	CFU/100 mL	0	N/A	45	33	N/A	2	0

* ECR 1997 (amended 2023) does not specify any standard for EC. The limit of 1,500 μS/cm is adopted from published studies on drinking water in Bangladesh and comparable settings [57, 58], and is used here as a permissible limit, not as a national standard.

The DSPs serve the largest dependent population (2,400 households across 12 villages), consistent with their role as the primary year-round water source where all other options are either seasonal or of compromised quality. CRWHs serve the smallest catchments (180 households across 7 villages; maximum access distance 2 km) because of constraints on storage volume and dependence on seasonal rainfall. The access distances for DTW, PSF, SPSF, DSP, and UFP all reach or exceed 4 km for peripheral villages, reflecting the low density of installed units relative to demand. DSP is the only technology used year-round without seasonal restrictions, whereas other DWTs are confined primarily to the dry season, when surface water alternatives are most depleted.

4.3 Water quality assessment

The results of the selected water quality parameters are summarised in Table 2, and the scoring of the water quality supplied by each DWT is provided in the Supplementary Material (Table S4). All DWTs produced water within the ECR permissible pH range (6.5–8.5), except CRWH, which slightly crossed the upper limit. Values are means of three units per technology (two for UFP) sampled once during the peak dry season and should be read as indicative of between-technology differences rather than as technology-wide characterisations.

EC, representing salinity, reveals the most critical quality difference. DTW exhibited extreme salinity (EC = 2,601 μS/cm; TDS = 1,302 mg/L), exceeding both the adopted EC limit and the ECR limit for TDS, and confirming that groundwater is intrinsically unsuitable for potable use in this geological setting. SPSF water also exceeded the EC limit (1,746 μS/cm), reflecting the accumulation of dissolved salts as source pond salinity increased during the peak dry season. In contrast, CRWH presented the lowest salinity of all DWTs (EC = 121 μS/cm; TDS = 60 mg/L). DSP achieved effective salinity removal (EC = 378 μS/cm), as did UFP (EC = 748 μS/cm), whereas PSF remained within the limit

while recording the third highest salinity (1,202 $\mu\text{S}/\text{cm}$). The EC and TDS profiles are shown in the Supplementary Material (Figs. S6, S7).

Microbiological quality, assessed through FC counts, revealed severe contamination risks in surface water-based DWTs, with the exception of CRWH. PSF presented the highest FC count (45 colony-forming units (CFU)/100 mL), followed by SPSF (33 CFU/100 mL), both of which far exceeded the ECR standard of zero detectable coliforms in 100 mL. UFP showed minimal but non-zero contamination (2 CFU/100 mL), which is indicative of post-filtration recontamination during storage or distribution. CRWH uniquely achieved a complete absence of FC (0 CFU/100 mL), which was attributable to its closed collection and storage system. The FC profiles are presented in the Supplementary Material (Fig. S8). Nutrient contamination (TN, TP) exceeded the ECR thresholds across most surface water-based DWTs. SPSF showed the most severe TN exceedance (25.7 mg/L vs. the standard of ≤ 10 mg/L), reflecting intensive nutrient loading from pond water receiving domestic wastewater and agricultural runoff. CRWH marginally exceeded both the TN standards (13.0 mg/L) and the TP standards (0.050 mg/L). PSF and UFP exceeded both the TN and TP standards; all exceedances were attributable to nutrient-enriched source water. The TN and TP profiles are shown in the Supplementary Material (Fig. S9).

4.4 Climate resiliency of DWTs

4.4.1 Robustness aspects

The structural resistance of DWTs to climate hazards varies considerably across technology types. The study area faces cyclonic winds, embankment failures, storm surges, floods, heatwaves, and droughts, with embankment failures triggered by cyclonic winds and storm surges identified as the most destructive (see Supplementary Material, Fig. S10). DSP has the highest structural resistance owing to its flood-proof construction and reliance on protected groundwater sources (Fig. 4). CRWH, despite its large exposed concrete tank susceptible to high winds, demonstrates strong extrinsic resiliency through active community-based maintenance, enabling rapid recovery even after physical damage [59]. DTW lacks flood-proofing and is exposed during inundation, whereas

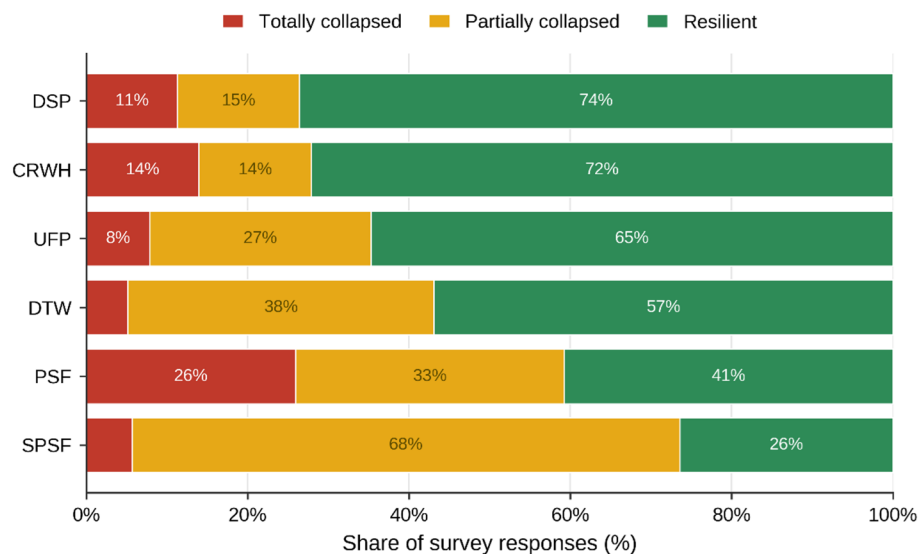


Fig. 4 Comparative disaster resiliency of different DWTs (N = 180)

PSF and SPSF are the most structurally vulnerable, with exposed storage structures prone to damage and filter blockage during extreme weather, including heatwave-driven demand spikes that overwhelm slow sand filtration capacity.

Household experience of failure supports this ordering. In the case of failure of DWT during its operating life, 90% of SPSF users and 70% of PSF users reported at least one failure, compared with 47% for DSP, 27% for DTW, 20% for UFP, and 13% for CRWH. The contrast between CRWH and the pond filtration technologies is the widest on any single indicator in the study. From the perspective of water quality, DSP performs best on the parameters measured for it, followed by CRWH, whose only shortfall is a marginal TN exceedance. UFP maintains an acceptable EC (748 $\mu\text{S}/\text{cm}$) but shows low-level faecal contamination and elevated nutrients. DTW is constrained by extreme salinity (EC = 2,601 $\mu\text{S}/\text{cm}$; TDS = 1,302 mg/L), indicating groundwater unsuitability. SPSF and PSF were the lowest due to elevated salinity, the presence of faecal contamination (33–45 CFU/100 mL), and the highest nutrient concentrations.

Operating life was scored from the design lives reported by implementing agencies and owners during the KIIs, cross-checked against respondent estimates. On the design-life basis, CRWH leads at approximately 50 years, which is attributable to durable reinforced concrete construction and the low operational stress of gravity-fed collection. DSP averages approximately 30 years, SPSF around 25 years, and PSF and UFP approximately 20 years. DTW has the shortest design life at approximately 15 years, owing to saline corrosion of its mechanical components. Design lives rather than respondent estimates were therefore used for scoring, and both are reported here so the basis of the judgement is transparent.

4.4.2 Redundancy aspects

Source substitutability differs markedly across DWT types. Approximately 70% of respondents rely on household-based rainwater harvesting during the rainy season as their primary source, reflecting its reliability and minimal cost; however, household harvested rainwater is typically depleted within 6–8 months, after which households shift to alternative technologies (Fig. 5), with DSP serving as the sole year-round provider. This is the basis of the source substitutability scores: DSP alone can absorb demand displaced

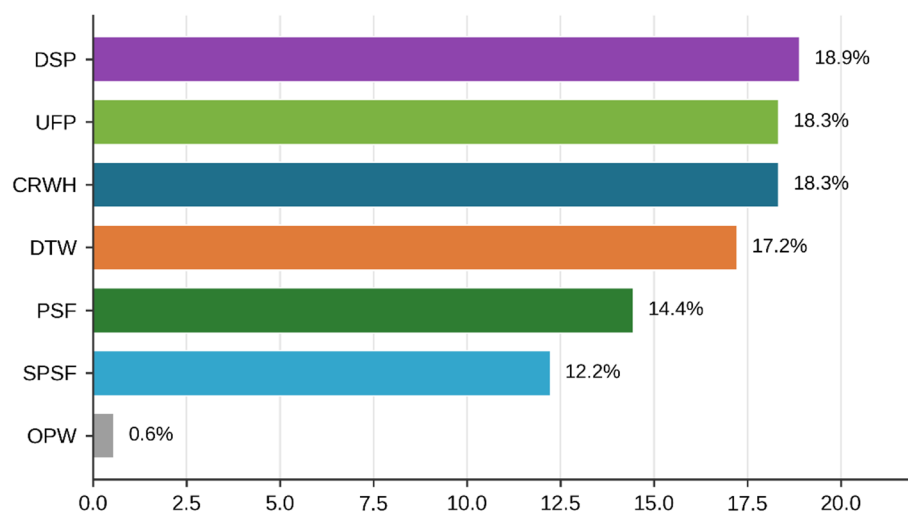


Fig. 5 Household dependency on DWTs during the dry period (N = 180)

from a failed or exhausted source throughout the year and therefore scores highest; PSF and SPSF operate year-round but with declining and increasingly saline output. CRWH, DTW, and UFP cannot substitute for a failed source in the late dry season, whether because storage is exhausted (CRWH), the water is unusable for drinking (DTW), or coverage is limited (UFP).

Management committees, the designated institutional support system for community DWT maintenance, are broadly inactive across most technology types, a pattern consistent with recent multi-country evidence including Bangladesh [31, 60]. Only CRWH departs from this pattern: 63% of CRWH users reported an active local committee, against 7% or fewer for every other technology (PSF and UFP; DSP, SPSE, and DTW). CRWH committees are linked to mosques and actively monitor and maintain their units, enabling prompt problem-solving without external support. Committees for SPSE, PSF, and DTW remain inactive for want of maintenance financing; members express willingness to work, but their efforts depend on resources that implementing agencies no longer provide after project completion. DSP and UFP operate under owner-driven models that ensure individual accountability but exclude community governance entirely, mirroring Hoque et al.'s [61] finding that the absence of post-installation institutional anchoring drives recurrent water-system failure.

Spare-part supply follows the same divide. Asked whether spare parts were available in the local market, 97% of DTW users and 90% of CRWH users answered yes, against 40% for SPSE, 23% for PSF, 17% for DSP, and 3% for UFP. For the four latter technologies, the largest response category was 'don't know' (15–23 respondents each), which is itself an indicator of how far the supply chain sits outside community knowledge. Simple components for CRWH, DTW, PSF, and SPSE, such as pipes, motors, taps, and solar panels, are stocked in local markets, whereas DSP and UFP owners must source membranes and pressure vessels from the nearest cities (e.g., Khulna or Satkhira), incurring procurement lead times that delay restoration when it is most needed (see Supplementary Material, Figs. S11–S12).

4.4.3 Resourcefulness aspects

The role of gender in management reveals further disparities. Although implementing agencies conduct pre-installation consultations involving both men and women, female voices remain constrained by social norms in mixed-gender forums (see Supplementary Material, Fig. S13). Among the 74 respondents who recalled such a consultation, 59 reported that women participated, and of those, 57 believed women's concerns were considered; the constraint therefore operates less through formal exclusion than through the translation of participation into sustained influence. CRWH committees are male-led, while PSF, SPSE, and DTW committees formally include women but are inactive. An exceptional case is the community-driven DSPs funded by the Social Development Foundation (SDF), managed exclusively by women's committees whose savings financed minor repairs. However, all such units became non-functional once maintenance costs exceeded the committee's financial capacity, demonstrating that gender-inclusive governance requires adequate financial backing to be sustainable.

Maintenance frequency and repair capacity separate the technologies further. Reported maintenance requirements are heaviest for the pond-based and membrane systems: 90% of SPSF users and 70% of PSF users could state a routine maintenance

interval, most commonly monthly or more frequent, and DSP owners reported the most frequent servicing of all, with weekly attention cited most often among the 14 who specified an interval. By contrast, only 4 out of 30 CRWH users could identify any routine maintenance requirement beyond periodic tank cleaning, typically annual. This maps directly onto cost. DSP and UFP incur the highest annual O&M expenditure at about BDT 50,000 per year each, because of imported chemicals and components that require external technicians. CRWH, PSF, and DTW remain the most affordable at BDT 1,000–2,000 per year, since maintenance consists largely of routine cleaning.

Ease of repair, measured by the availability of skilled technicians, also shapes functionality (Fig. 6). Local repair capacity is reported by 93% of DTW users and 90% of CRWH users, falling to 37% for PSF, 23% for SPSF, 17% for DSP, and 10% for UFP. CRWH and DTW faults are consequently resolved by local technicians within a day or two, whereas PSF and SPSF repairs take considerably longer because of technical complexity combined with dependence on committee-organised funding. Notably, no formal technical training exists for local DWT repair, leaving communities structurally dependent on external expertise for the advanced systems.

4.4.4 Rapidity aspects

Most DWTs maintain physical functionality during disasters, with the notable exception of PSF and SPSF. Almost 50% of PSF and SPSF users reported operational disruptions due to pond contamination from domestic wastewater, agricultural runoff, embankment failures, and storm surges. Energy-dependent systems face a compounding indirect vulnerability: DSP, UFP, and SPSF become inoperative during and after disasters because of electricity and solar energy disruptions. Even when their physical structures remain intact, pre-disaster load-shedding and post-cyclone power outages coincide with peak safe-water demand. Access, rather than the installation itself, is the more widespread constraint: 73% of respondents reported difficulty reaching water sources after disasters, primarily due to inundation and damaged access routes, muddy and slippery roads, or blockage caused by uprooted or broken branches of trees (Fig. 7). Differences in accessibility during extreme events nevertheless follow the same pattern as the other rapidity

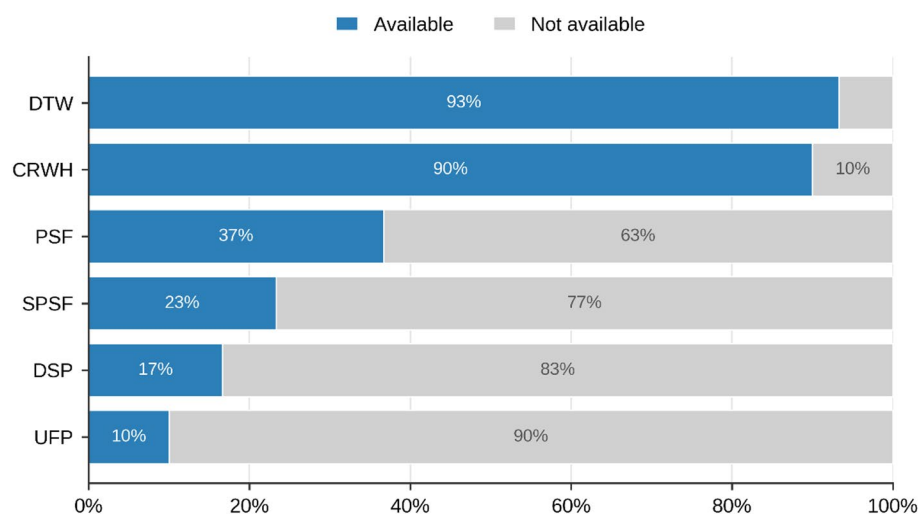


Fig. 6 Availability of skilled technicians for DWT repair (N=180)

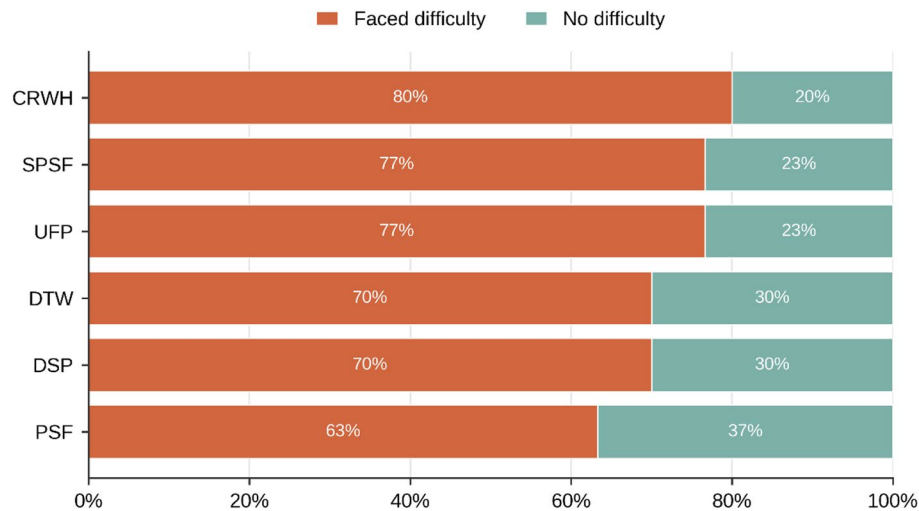


Fig. 7 Respondents reporting difficulty reaching DWTs after disasters (N=180)

indicators. DTW was cited as accessible by 83% of its users and CRWH by 73%, against 57% for UFP, 53% for SPSE, 50% for PSE, and 47% for DSP.

Emergency response committees are uncommon except for CRWH, reported by 53% of its users, and SPSF, reported by 50%; for the remaining technologies, the figure ranges from 3% to 17%. Reported restoration times mirror this institutional readiness. CRWH achieves the fastest average emergency repair time of 1.3 days, facilitated by mosque-linked committees with pre-established mobilization capacity. DTW is also repaired within 1 day, because spare parts and technicians are simple and readily available. DSP is restored within 2.5 days under owner-driven action and UFP within five days and SPSF within 6.6 days, the delay reflecting the time taken to collect repair funds from beneficiaries of the free service. PSF has the longest average repair time, at 11.4 days, reflecting both technical complexity and reliance on committee-based fund mobilisation. Access to DWTs during and after disasters and the speed of emergency response are critical determinants of community resiliency. Social support for restoration varies by ownership model. For free-service DWTs (PSF, SPSF), maintenance costs fall on users, who express a strong willingness to contribute labour and funds. However, the time required to mobilise these resources translates into extended downtime. CRWH benefits from a hybrid funding model that combines agency support, community contributions, and mosque-institutional backing. DSP and UFP, as paid-service systems, depend entirely on owners, limiting collective community solidarity in post-disaster restoration. A critical cross-cutting capacity gap is also evident. In total, 54% of respondents reported an inability to assist with O&M, and only 2 of the 180 had ever received any training in operating or maintaining DWT. This highlights the urgent need for structured technical training programmes to build community-level repair competence across all DWT types (see Supplementary Material, Fig. S14).

4.4.5 Resiliency ranking of DWTs

The climate resiliency of each DWT was quantified via the WSM under the 4R framework, building on the evidence presented above. As described in Sect. 3.5, each of the twelve criteria was scored 1 (low), 2 (moderate), or 3 (high). Raw scores were normalised for each criterion (Eq. 1) and multiplied with equal weights of 0.083 (Eq. 2), so that the

Table 3 MCA for the climate resiliency of the six DWTs in the study area

4R dimensions	Criterion	Weight	CRWH	DTW	PSF	SPSF	DSP	UFP
Robustness	Water Quality	0.083	0.021	0.014	0.007	0.007	0.021	0.014
	Operating Life	0.083	0.025	0.008	0.008	0.017	0.017	0.008
	Structural Resistance	0.083	0.021	0.014	0.007	0.007	0.021	0.014
Redundancy	Source Substitutability	0.083	0.008	0.008	0.017	0.017	0.025	0.008
	Spare Parts Availability	0.083	0.021	0.021	0.014	0.014	0.007	0.007
	Support System	0.083	0.025	0.008	0.008	0.008	0.017	0.017
Resourcefulness	Gender Role in Management	0.083	0.021	0.007	0.021	0.021	0.007	0.007
	Operation & Maintenance	0.083	0.023	0.008	0.015	0.023	0.008	0.008
	Ease of Repair	0.083	0.021	0.021	0.014	0.014	0.007	0.007
Rapidity	Accessibility During Disasters	0.083	0.023	0.015	0.008	0.015	0.008	0.015
	Emergency Response	0.083	0.023	0.008	0.008	0.015	0.015	0.015
	Social Support	0.083	0.015	0.008	0.023	0.023	0.008	0.008
Weighted total score		1.000	0.246	0.139	0.149	0.180	0.159	0.127
Resiliency rank		—	1st	5th	4th	2nd	3rd	6th

Bold values indicate the weighted total scores and corresponding resiliency ranks of the DWTs

composite score of a technology expresses its share of the total assessed 4R performance of the six-technology set, against an equal-share benchmark of 0.167. The complete scoring matrix and final weighted scores are presented in Table 3, with the full calculation provided in the Supplementary Material (Tables S4 – S6).

The MCA produced a final ranking of CRWH (0.246) > SPSF (0.180) > DSP (0.159) > PSF (0.149) > DTW (0.139) > UFP (0.127). CRWH attained the highest share, 48% above the equal-share benchmark and 1.9 times that of the lowest-ranked technology, reflecting the combined advantages of community governance, active maintenance, and long operational life. SPSF ranks second, scoring well on gender-inclusive management and social support, consistent with its community-facing free-service model and moderate local maintenance capacity, though constrained by low structural resistance and solar-energy dependency. DSP ranks third despite delivering the best measured water quality and the widest year-round coverage, because its owner-driven model and dependence on imported-technology depress its scores for spare-part availability, ease of repair, and social support. This is the study's central tension in numerical form: the technology most depended upon for year-round supply is also among the most energy-vulnerable when supply matters most.

PSF ranks fourth: it scores comparably to CRWH and SPSF on gender roles and social support, reflecting community willingness, but its low water quality, structural resistance, and support system scores reflect governance and maintenance failure. DTW ranks fifth, penalised principally by water quality and emergency response, since the technology remains functionally unusable for drinking even when mechanically operational. UFP ranks last (0.127), reflecting the compounding of energy dependency, imported spare parts, high O&M costs, paid-service exclusion, and the absence of community governance across all four dimensions. Overall, technologies combining community participation, local resource availability, and low external dependency achieve higher shares, whereas those relying on centralised ownership, imported components, and external energy score systematically lower.

5 Discussion

In this study, the climate resiliency of DWTs was governed less by technological sophistication or water quality than by institutional governance, community ownership, energy independence, and local maintenance capacity. This is evident from the contrasting rankings produced by the same six technologies. Based on the 4R resiliency assessment, the ranking was CRWH > SPSF > DSP > PSF > DTW > UFP, whereas the water quality ranking was DSP > CRWH > UFP > DTW > SPSF > PSF. Only CRWH ranks highly on both. This is the pattern Butler et al. [28] anticipate in arguing that dependable water management requires the TOSE dimensions to be integrated rather than traded off, and it is consistent with Charles et al.'s [31] finding that technological complexity without commensurate local capacity produces poorer long-term outcomes. The 50% non-functionality rate observed here sits at the upper bound of the 30–50% range reported across South Asia and sub-Saharan Africa [20, 21]. This highlights a persistent hardware–software investment gap, where infrastructure is installed without concurrent development of management capacity, maintenance financing, or accountability mechanisms [36, 62].

Water quality, reported in Sect. 4.3, reflects the condition of the water source and the effectiveness of the treatment process at a specific point in time. For example, membrane-based desalination removes salinity that filtration technologies cannot [63], closed rainwater harvesting prevents the faecal contamination pathways common to open ponds [64], and pond-fed filters inevitably reflect the quality of their source water. Resiliency, by contrast, is a property of the socio-technical system that sustains service delivery over time. It depends on who repairs the system, the availability of spare parts, financing arrangements, and the speed of maintenance. Consequently, water quality and resiliency are only loosely coupled, allowing a technology to perform well on one measure yet rank lower on the other. This divergence is not an artefact of the assessment framework but its central finding. It also has an important policy implication: procurement decisions based solely on water quality are likely to overvalue technologies that depend on external energy sources and imported components.

Three mechanisms, rather than differences among the six technologies themselves, explain most of the variation in the resiliency ranking. The first is energy dependence. Technologies relying on grid or solar power (DSP, UFP, and SPSF) are most vulnerable during periods of peak demand because Bangladesh's coastal electricity grid often experiences multi-week outages after major cyclones [65], while post-storm cloud cover reduces solar power generation [23, 61]. In contrast, gravity-fed CRWH and manually operated DTW carry no such exposure. The second mechanism is the length and locality of the repair chain. Where spare parts and technical expertise are available locally, systems are typically restored within a day. However, technologies requiring membranes, pressure vessels, and external technicians from nearby cities such as Khulna or Satkhira often remain non-functional for several days or even weeks. The third is the institutional anchor. CRWH is the only technology managed through a pre-existing and trusted local institution, the mosque, rather than a committee established specifically for the project. It is also the only technology with majority-reported active committee functioning and the shortest recovery time. This finding is consistent with Kim et al. [59], who identified the social embeddedness of water committees as the strongest cross-national predictor of maintenance response across 23 Asian and African countries.

These mechanisms operate independently of the conventional low-technology to high-technology axis and explain the apparent paradoxes in the ranking. DSP ranks third rather than first because its excellent water quality and widest coverage cannot compensate for a repair chain that runs through another district and a power supply that fails with the storm. SPSF ranks second rather than fourth because strong community governance and a free-service model partly offset its structural fragility. By contrast, UFP ranks last because all three mechanisms act against it simultaneously. CRWH performs best not because rainwater is technically superior, but because it performs well across all three mechanisms. Its principal weakness lies elsewhere, namely limited storage capacity, which is typically exhausted within 6–8 months after the monsoon ends. The 4R criteria capture this limitation only indirectly, indicating that storage capacity remains an important dimension in policy decisions.

Gender shapes exposure to water insecurity more strongly than technology preference. Women constitute 55% of water collectors in the study area and absorb the additional distance and risk that follow a disruption. However, their formal influence over water management decisions remains constrained by prevailing social norms. This pattern has been widely documented in coastal Bangladesh, where gender roles assign women responsibility for daily water collection while limiting their participation in formal water governance [66]. The study also found that most respondents who recalled a pre-installation consultation reported that women participated and that their concerns were considered. The constraint therefore lies less in access to the consultation process than in sustaining women's influence after project implementation. The all-female DSP committees established under the SDF illustrate both the potential and the limitation of gender-inclusive governance. These committees functioned effectively until maintenance costs exceeded their financial capacity, after which every unit became non-functional. Gender-responsive governance in this setting requires adequate financial resources, not merely women's representation in management committees.

Affordability operates along a separate dimension and creates a distinct equity problem. Technology use is determined primarily by proximity rather than household choice. As a result, the cost of drinking water is effectively determined by location. Every UFP user and 87% of DSP users pay for drinking water, compared with only 13–17% of PSF and SPSF users. Moreover, 40% of paying households reported reducing other household expenditure to meet these costs. Paid-service models therefore impose the greatest recurring financial burden on households that had no influence over the technology installed nearby. In contrast, free-service technologies supported by active community governance are better aligned with the circumstances of marginalised populations [67]. Although this distributional issue is distinct from the resiliency ranking, it reinforces the conclusion that technology selection without consideration of the ownership model shifts risk onto those least able to bear it.

Comparable dynamics have been reported in other salinity-affected deltaic and island settings. Previous research on delta water management shows that adaptation to increasing salinity depends more on effective governance and anticipatory planning than on additional infrastructure [68]. In areas where rainwater harvesting is the primary source of drinking water, the main constraints are adequate storage through the dry season and the institutional capacity to maintain distributed systems [69]. Similar conclusions have been drawn across multiple countries by Charles et al. [31]. Likewise, Whittington et al.

[40] attribute rural water system failures primarily to governance rather than technology. Harvey and Reed [62] further demonstrate that community-managed water supplies fail when post-installation financing is absent, a pattern also reported for pond-based water systems in this region [70]. The convergence suggests that the mechanisms identified here describe a recognisable failure mode in coastal water provision, though how far the specific ranking would transfer remains untested.

Climate change is likely to intensify these constraints. Continued sea-level rise and reduced upstream flow during the dry season will increase salinity intrusion [5, 6, 71, 72], further reducing the viability of DTW and degrading the source ponds used by PSF and SPSF. At the same time, longer dry seasons [65] will accelerate the depletion of CRWH storage and increase reliance on energy-dependent technologies that perform poorly during disruptions. These projected changes reinforce the central finding of this study that long-term climate resiliency depends not only on technical performance but also on governance, energy security, and local maintenance capacity. The 4R framework, operationalised through MCA, provides a transparent approach for evaluating these interdependent dimensions and offers a practical basis for more resilient drinking water planning in climate-vulnerable coastal settings.

6 Conclusions

This study assessed the climate resiliency of six co-existing DWTs in a hard-to-reach community of south-western coastal Bangladesh using the 4R framework and MCA. The principal conclusion is conceptual rather than technological. Water quality and climate resiliency are empirically distinct properties of DWTs. Technology that delivers excellent-quality water may still fail when safe water is most needed. Differences in resiliency are explained primarily by energy dependence, the length and locality of the repair chain, and the presence or absence of an institutional anchor, rather than by technological sophistication.

These findings have three practical implications. First, DPHE and implementing NGOs should assess the 4R criteria alongside technical suitability when selecting DWTs. Capital investment should be linked to a maintenance financing plan, with a revolving maintenance fund established at project inception through a fixed share of installation costs and managed by the Union Parishad. Second, because no single technology can provide reliable service throughout both the wet and late dry seasons, investment should prioritise complementary technology portfolios at the catchment level rather than a single highest-ranked option. Retrofitting existing plants with solar-battery systems offers a practical approach to securing year-round service. Third, local repair capacity should be strengthened through structured technician training programmes implemented with LGED and NGO partners, with explicit provision for women's participation.

Future research should assess whether the resiliency mechanisms identified in this study operate similarly across other salinity-affected coastal settings. Longitudinal monitoring, coordinated with Union Parishad records, would track how performance and governance shift as salinity intrusion and dry seasons lengthen. At the planning stage, participatory assessments should guide government and NGOs in matching technologies to local capacity and affordability. Pilot projects should also evaluate Union Parishad-managed maintenance funds, solar battery retrofits for energy-dependent systems, and technician training with explicit provision for women's participation before scale-up.

Supplementary Information

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Supplementary Material 1

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Author contributions

SK: Conceptualisation, methodology, field data collection, formal analysis, visualisation, writing—original draft. MSM: Conceptualisation, supervision, methodology, writing—review and editing. MM: Conceptualisation, field data collection, formal analysis, writing—review and editing. All the authors read and approved the final manuscript.

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Data availability

The datasets generated and analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval

This study was approved by the Research and Academic Committee (RAC) of the Institute of Water and Flood Management (IWMF) and the Committee for Advanced Studies and Research (CASR) of the Bangladesh University of Engineering and Technology (BUET). Ethical approval was subsequently granted by the Institutional Review Board (IRB), BUET (Approval No. BUET/RISE/IRB/2026-03-001; 04 March 2026). An earlier ethical clearance had also been obtained from the Ethical Review Committee of the University of Peradeniya (Approval No. ECC/2023/R/68; 29 January 2024). The study was conducted in accordance with the guidelines of the BUET IRB and the principles of the Declaration of Helsinki. There was no experiment in the study involving humans, and there are no personally identifiable data in the manuscript.

Consent for publication

Not applicable.

Informed consent

Verbal informed consent was obtained from all survey respondents and FGD/KII participants before data collection. Participation was voluntary, and anonymity was maintained throughout the analysis and reporting.

Competing interests

The authors declare no competing interests.

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